

Verification deck for parameter-response falsification

The deck assumes you do not want to line-by-line audit every implementation detail. Instead, you audit each structure and each interface by sweeping the knobs that should expose incorrect behavior first.

Report Order

- Four structures: parameters and expected response
- Six pairwise interfaces: direct or mediated
- Expected behavior versus actual behavior slot
- End with a repeatable execution checklist

Core matrix views in this study:
reduced $Y_{dm,in}(\omega)$, full $Y(\omega)$, $Z \rightarrow Y$
reductions, and readout-branch S21
with vector fitting.

Three-Part Verification Logic

This deck is intentionally about verification first, solution verification second, and validation third.

1. Verification

Did the code implement the intended mathematical structure? Use parameter-response checks and decoupling limits.

2. Solution Verification

Are discretization, sweep resolution, matching windows, and fitting settings converged enough for this claim?

3. Validation

Do the trends and magnitudes still look like plausible physics, not only plausible numerics?

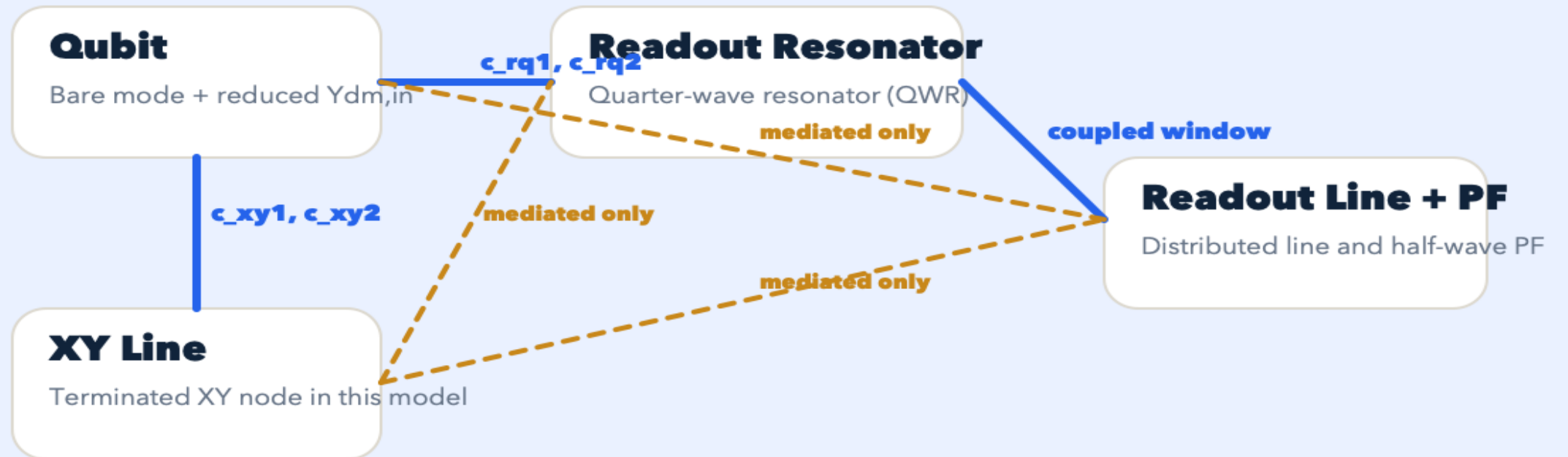
Operational Rule For This Study

- Start with self-structure sweeps where the expected trend is simplest and strongest.
- Move to direct interfaces and check monotonicity under the explicit coupling knob.
- Treat the remaining three pairs as mediated only, and audit them with path-breaking null tests.

Model Topology Used By The Audit

This summary matters because some interfaces are direct in the netlist, while others are only mediated through an intermediate path.

Four structures, three direct interfaces, three mediated interfaces



Direct interfaces should be tested with monotonic parameter sweeps. Mediated interfaces should be tested with path-breaking null che

Pairwise Audit Inventory

Direct interfaces get sweep-based checks. Mediated interfaces get null checks and path-breaking checks.

Upper triangle shows how each pair should be audited

	Qubit	Readout Resonator	XY Line	Readout Line + PF
Qubit		Direct	Direct	Mediated
Readout Resonator			Mediated	Direct
XY Line				Mediated

Direct pairs get sweep-based monotonic checks. Mediated pairs get null tests and path-breaking checks.

Qubit

Treat the qubit as the reference oscillator. This is where monotonicity checks should be easiest and where nonphysical loss should be easiest to spot.

Knobs In This Study

l_{q_h}
 c_{q_f}
 c_{g1_f}, c_{g2_f}
 $qubit_port_res_ohm$

Primary Matrix View

Primary observable: reduced $Y_{dm,in}(\omega)$,
 extracted f_q from $\text{Im}(Y_{dm,in})=0$, and
 $\text{Re}(Y_{dm,in})$ at resonance.

Expected Qualitative Response

- L_q up should move f_q down.
- C_q or pad-to-ground capacitance up should move f_q down.
- With $c_{xy} = c_{rq} = 0$, $\text{Re}(Y_{dm,in})$ should stay small and smooth.
- Pad asymmetry should affect CM/DM weighting, not spawn unrelated resonances.

What Counts As A Good Sign

- Zero crossing moves monotonically under parameter sweeps.
- Retuning the bare qubit does not suddenly create large loss.
- Frequency extraction is stable under sweep refinement.

Actual behavior / plot slot

Insert: L_q sweep, $\text{Im}(Y_{dm,in})$ zero-crossing traces, and one isolated-qubit sanity plot.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

Readout Resonator

In this study the readout resonator is the quarter-wave resonator branch coupled into the readout line through the coupled window.

Knobs In This Study

qwr_length_m
common_l_per_m_h, common_c_per_m_f
common_r_per_m_ohm, common_g_per_m_s
qwr_target_dz_m

Primary Matrix View

Primary observable: readout-line S21, fitted resonator poles from vector fitting, and optional local Y before reduction.

Expected Qualitative Response

- Longer electrical length should move the readout-resonator resonance lower.
- Changing common RLGC should shift distributed resonances consistently, not arbitrarily.
- If $c_{rq1} = c_{rq2} = 0$, retuning the resonator should not move the qubit result.
- Mesh refinement should converge mode locations instead of reordering them.

What Counts As A Good Sign

- The narrower fitted resonance tracks the readout resonator consistently.
- Resonance motion follows electrical length intuition.
- The resonator disappears from qubit behavior when the qubit interface is opened.

Actual behavior / plot slot

Insert: qwr length sweep on S21 and fitted resonance markers with c_{rq} toggled on/off.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

XY Line

Important modeling fact: in this study the XY line is represented as a terminated XY node plus the qubit-to-XY coupling capacitors, not as a long distributed geometry block.

Knobs In This Study

`c_xy1_f`, `c_xy2_f`
`xy_port_res_ohm`
 common RLGC only matters indirectly here
 No independent XY length parameter in this model

Primary Matrix View

Primary observable: port-3 loading in the full $Y(\omega)$ and its effect on reduced $Y_{dm,in}(\omega)$.

Expected Qualitative Response

- Increasing `c_xy` should strengthen the XY-only loading contribution.
- Setting `c_xy1` = `c_xy2` = 0 should collapse the XY contribution cleanly.
- Changing `xy_port_res_ohm` should alter dissipative loading but should not create readout resonances.
- Any claim about XY distributed standing-wave motion should be treated carefully because this model does not expose an XY length knob.

What Counts As A Good Sign

- XY-only loss vanishes in the zero-coupling limit.
- XY trends are monotonic with `c_xy` sweeps.
- The study narrative stays aligned with what the model actually contains.

Actual behavior / plot slot

Insert: `c_xy` sweep, `xy_port_res` sweep, and one note explaining the XY node abstraction.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

Readout Line With Purcell Filter

This is the distributed branch from readout input to output, including left line, half-wave Purcell filter, right line, and terminations.

Knobs In This Study

left_readout_length_m,
right_readout_length_m
purcell_filter_length_m
pf_coupling_cap_in_f,
pf_coupling_cap_out_f
readout_port_res_ohm
left/right/pf_target_dz_m

Primary Matrix View

Primary observable: raw readout-line S21,
fitted S21 model, and extracted Purcell-filter /
readout-resonator resonances.

Expected Qualitative Response

- Increasing electrical length should move the associated resonances or notches lower.
- Changing PF coupling capacitors should alter bandwidth and feature strength.
- Port termination changes should alter line response smoothly, not invent extra physics.
- Discretization refinement should stabilize S21 features and fitted poles.

What Counts As A Good Sign

- The broader fitted feature tracks the Purcell-filter mode consistently.
- Mode motion is consistent with length scaling and coupling strength.
- The branch can be characterized cleanly even when qubit couplings are disabled.

Actual behavior / plot slot

Insert: S21 sweeps versus PF length and coupling-cap sweeps, plus vector-fitting overlays.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

Qubit ↔ XY Line

Each pair should be audited by asking whether the parameter-response pattern matches the topology that this model actually contains.

Controlling Knobs

`c_xy1_f`
`c_xy2_f`
`xy_port_res_ohm`

Primary Matrix View

Use reduced $Y_{dm,in}(\omega)$ near qubit resonance plus selected full $Y(\omega)$ traces for port 3.

Expected Behavior

- Increasing c_{xy} should increase XY-only loading and usually increase $\text{Re}(Y_{dm,in})$.
- c_{xy} imbalance changes the centroid weights w_1 and w_2 , so asymmetry should show up systematically.
- Setting $c_{xy1} = c_{xy2} = 0$ should collapse to the no-XY reference.

If This Fails, Suspect

- Loss stays large even when c_{xy} is zero.
- f_q moves in the opposite direction with stronger XY loading without another explanation.
- Random extra resonances appear when only c_{xy} is swept.

Actual behavior / plot slot

Insert: XY-only sweep panel with c_{xy} magnitude and imbalance variants.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

Qubit ↔ Readout Resonator

Each pair should be audited by asking whether the parameter-response pattern matches the topology that this model actually contains.

Controlling Knobs

`c_rq1_f`
`c_rq2_f`
`qwr_length_m` (alignment context)

Primary Matrix View

Use reduced $Y_{dm,in}(\omega)$ at the qubit plus S21 context from the readout branch.

Expected Behavior

- Increasing `c_rq` should strengthen readout-mediated frequency shift and loss.
- Tuning the resonator closer to the qubit should increase loading.
- Setting `c_rq1 = c_rq2 = 0` should disconnect the qubit from the entire readout branch.

If This Fails, Suspect

- The qubit still feels the readout branch when `c_rq` is zero.
- Changing `qwr` length strongly affects the qubit with the interface opened.
- Observed shift and loss move incoherently under the same `c_rq` sweep.

Actual behavior / plot slot

Insert: `c_rq` sweep, `qwr` detuning sweep, and readout-off null check.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

Readout Resonator ↔ Readout Line + Purcell Filter

Each pair should be audited by asking whether the parameter-response pattern matches the topology that this model actually contains.

Controlling Knobs

`coupled_window_length_m`
`coupled_window_input_mode`
`q2d_*` or `modal even/odd parameters`
`pf_window_start_m`, `qwr_window_start_m`

Primary Matrix View

Use raw S21, fitted S21 model, and extracted resonances from the readout branch.

Expected Behavior

- Longer or stronger coupled-window interaction should increase hybridization.
- Window length $\rightarrow 0$ or mutual terms $\rightarrow 0$ should decouple the resonator from the line.
- Modal and q2d parameterizations should agree qualitatively on trend direction.

If This Fails, Suspect

- The resonator feature does not respond to window strength at all.
- The line still shows strong resonator loading with the window effectively off.
- q2d and modal modes disagree on basic trend direction.

Actual behavior / plot slot

Insert: coupled-window length sweep, modal-vs-q2d comparison, and one decoupling limit plot.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

Qubit ↔ Readout Line + Purcell Filter

Each pair should be audited by asking whether the parameter-response pattern matches the topology that this model actually contains.

Controlling Knobs

No direct knob in this model
Path toggles: `c_rq*`,
`coupled_window_length_m`, PF branch
lengths

Primary Matrix View

Use reduced $Y_{dm,in}(\omega)$ for the qubit and S_{21} for the readout branch as a path-consistency check.

Expected Behavior

- This effect must disappear if either the qubit↔resonator or resonator↔line interface is opened.
- Changing PF geometry with `c_rq = 0` should not move the qubit result.
- The effect should look path-mediated, not like a brand-new direct capacitor.

If This Fails, Suspect

- PF changes move the qubit even with `c_rq` opened.
- The qubit responds as if there were a direct line coupling that is absent from the netlist.
- Path-breaking toggles do not suppress the effect.

Actual behavior / plot slot

Insert: PF sweep with `c_rq` on/off, plus one path-break comparison panel.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

Readout Resonator $\leftrightarrow$ XY Line

Each pair should be audited by asking whether the parameter-response pattern matches the topology that this model actually contains.

Controlling Knobs

No direct knob in this model
Path toggles: c_{rq} , c_{xy}

Primary Matrix View

Use $Y_{dm,in}(\omega)$ and selected full-matrix traces only as a mediated-effect null test.

Expected Behavior

- This interaction should exist only through the shared qubit network.
- Zeroing either c_{rq} or c_{xy} should kill the path.
- Any residual feature should be small and systematic, not dominant.

If This Fails, Suspect

- A strong $RR \leftrightarrow XY$ effect remains after zeroing one side of the path.
- A new standalone feature appears that cannot be routed through the qubit.
- Trend magnitude is first-order when the topology only allows higher-order mediation.

Actual behavior / plot slot

Insert: c_{xy} / c_{rq} path-break matrix and one residual-effect summary.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

XY Line ↔ Readout Line + Purcell Filter

Each pair should be audited by asking whether the parameter-response pattern matches the topology that this model actually contains.

Controlling Knobs

No direct knob in this model
Path toggles: c_{xy} *, c_{rq} *,
 $\text{coupled_window_length_m}$

Primary Matrix View

Use $Y_{dm,in}(\omega)$ and branch S21 only as a mediated consistency check.

Expected Behavior

- This is a second-order path through the qubit and resonator network.
- Zeroing c_{xy} or c_{rq} should collapse the interaction.
- Changing PF geometry alone should not change the XY-only reference case.

If This Fails, Suspect

- PF geometry changes alter XY-only behavior without the readout path engaged.
- The mediated effect is larger than the direct interfaces without a clear explanation.
- Path-breaking toggles fail to suppress the effect.

Actual behavior / plot slot

Insert: XY-only versus XY+RO comparison with explicit path toggles.

Recommended inserts

- One sweep plot with a monotonic knob on the x-axis
- One null-test or decoupling-limit comparison
- One sentence on whether the observed trend matches expectation

How To Run The Audit

This last slide turns the deck into a repeatable protocol instead of a one-off presentation.

Per-Slide Fill-In Rule

- Name the parameter being swept and which structure or interface it belongs to.
- State the expected matrix signature before showing the plot.
- Add the actual plot and one sentence: matched / partially matched / failed.
- If it fails, point to the first plausible implementation surface to inspect.

Recommended Score Columns

```
structure_or_pair  
swept_knob  
primary_matrix_view  
expected_trend  
observed_trend  
pass_status  
confidence_delta
```

Minimum Black-Box Test Set

- One monotonic self-sweep per structure
- One zero-coupling or decoupling-limit test per direct interface
- One path-breaking null test per mediated interface
- One discretization or sweep-resolution refinement check on the final claim
- One external-physics sanity check on magnitude or mode location

Decision Rule

Do not ask whether the implementation feels reasonable. Ask whether the necessary failure signatures stayed absent under the sweeps that should have exposed them first.